Lecture 0

CSE250 – Circuits and Electronics

INTRODUCTION TO CSE250

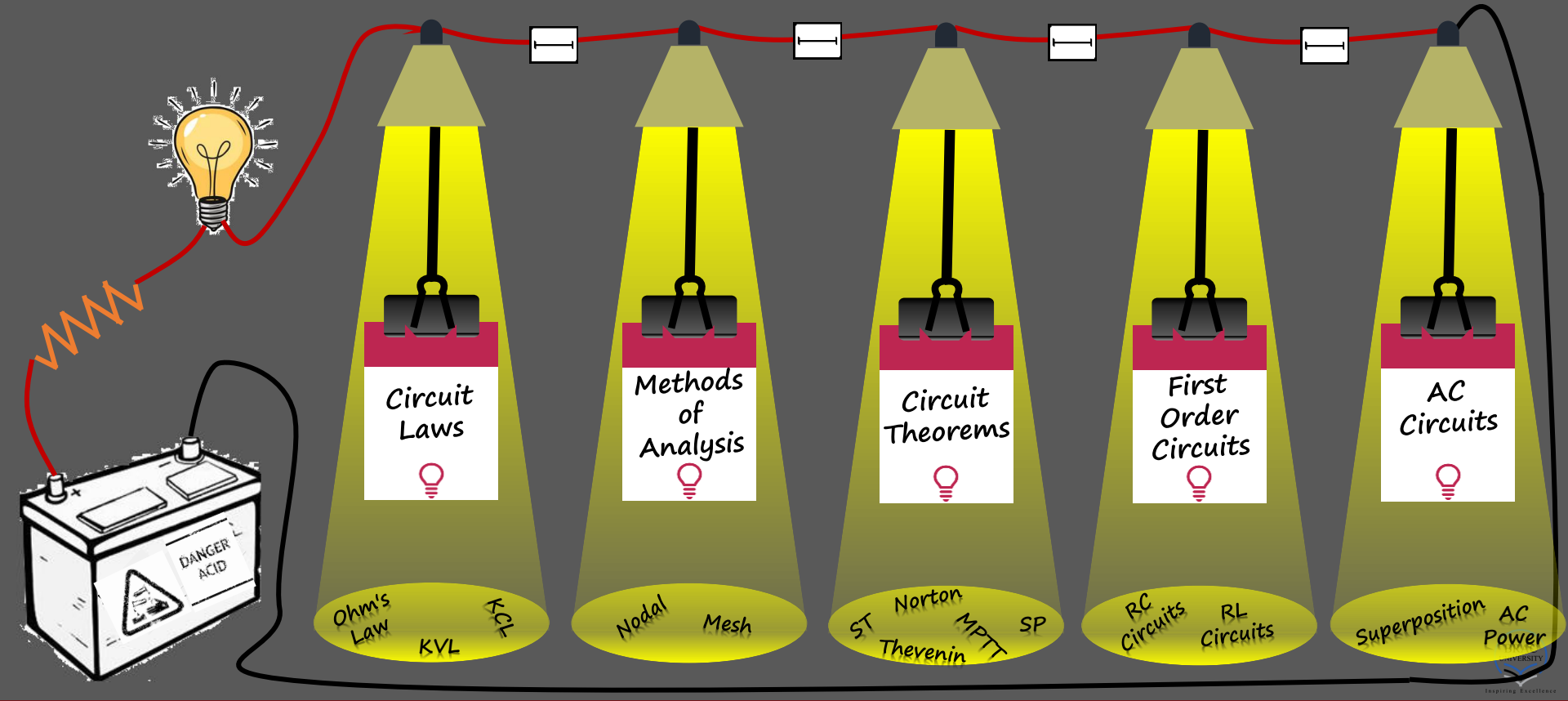


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Course Contents

This course provides a comprehensive introduction to linear electric circuit analysis, fundamental electrical concepts, and fundamental measuring units, combining theoretical mathematical modeling with mandatory weekly laboratory experimentation. The curriculum begins with the definition of basic electrical parameters (charge, voltage, current, energy, and power), the passive sign convention, and the physical I-V characteristics of ideal circuit components, including independent and dependent sources, resistors, capacitors, and inductors. Students examine fundamental governing laws, such as Ohm's law, Kirchhoff's current law (KCL), and Kirchhoff's voltage law (KVL), alongside circuit reduction methods including series and parallel equivalence and voltage and current divider rules. The course then covers systematic network analysis techniques, developing proficiency in nodal and mesh analysis for multi-loop direct current (DC) circuits containing both independent and dependent sources. Core linear circuit theorems, including superposition, source transformation, Thevenin and Norton equivalent circuits, and maximum power transfer, are analyzed to simplify complex electrical networks. Finally, the curriculum explores time-domain transient analysis of first-order RC and RL circuits, followed by alternating current (AC) steady-state circuit analysis using sinusoidal sources, phasor representations, and complex impedance concepts. A compulsory 3-hour weekly laboratory session reinforces theoretical concepts through hands-on circuit construction, measurement, and troubleshooting using standard equipment such as multimeters, oscilloscopes, and function generators.

Course Outline: broad themes



Course Objectives

a) Introduce fundamental electrical quantities (voltage, current, power), energy definitions, and the passive sign convention for electrical circuit analysis.

b) Illustrate the terminal I–V characteristics of two-terminal devices and ideal circuit components, including dependent and independent voltage and current sources, resistors, capacitors, and inductors—to infer circuit equivalence.

c) Explain governing circuit laws, including Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and Ohm's Law, alongside basic reduction techniques such as series–parallel equivalence and voltage–current divider rules.

d) Apply systematic circuit analysis techniques, specifically nodal analysis and mesh analysis, to solve multi-loop linear direct current (DC) circuits containing independent and dependent sources.

e) Demonstrate linear network theorems, including superposition, source transformation, Thevenin–Norton equivalent circuits, and the maximum power transfer theorem to simplify and analyze complex circuits.

f) Analyze the time-domain transient response of first–order RC and RL circuits, as well as the steady–state AC behavior of linear networks using phasor representation and complex impedance concepts.

g) Train students to construct, test, and troubleshoot linear electrical circuits in a weekly laboratory setting using standard electronic components.

h) Develop students' proficiency in using standard laboratory instruments—such as digital multimeters, oscilloscopes, and function generators—to measure and verify physical circuit parameters.

i) Enhance teamwork and technical communication skills through collaborative laboratory experimentation and structured technical report writing.

Course Outcomes

Upon successful completion of this course, you will be able to,

Sl.	CO Description	Weightage (%)
CO1	Apply foundational electrical concepts, governing circuit laws, and linear network theorems (superposition, source transformation, Thevenin, Norton, and maximum power transfer) to solve direct current circuits.	35
CO2	Analyze linear direct current, first-order transient (RC and RL), and alternating current circuits in the time and phasor domains using nodal analysis, mesh analysis, circuit equivalence, and voltage or current divider rules.	40
CO3	Function effectively as a member of a team to perform collaborative laboratory experiments and hardware setups.	7
CO4	Document and communicate experimental results, data analysis, and engineering insights clearly by preparing structured technical lab reports.	6
CO5	Collaborate effectively in a group in the laboratory, and **Report** their findings and insights clearly and concisely, using technical language and documentation standards.	12

Lesson Plan - 1/2

No	Topic	Lecture #
1	Course overview and motivation, real-world engineering applications of circuit theory, fundamental electrical units, charge, voltage, current, energy, and power definitions	Lecture 1
2	Passive sign convention, power calculation for active and passive elements, classification of circuit elements, independent and dependent DC voltage and current sources, standard circuit symbols	Lecture 2
3	Terminal I-V characteristics of basic two-terminal linear elements (resistors, independent voltage sources, independent current sources) and degenerate elements (open circuit and short circuit)	Lecture 3
4	Ohm's Law, basic circuit terminologies, series and parallel resistor combinations, concept of circuit equivalence, equivalent resistance calculation, handling open and short circuits	Lecture 4
5	Review of two-terminal I-V characteristics, real versus ideal sources, Source Transformation Theorem, problem-solving applications for circuit simplification	Lecture 5
6	Kirchhoff's Current Law (KCL) statements and applications, Current Divider Rule (CDR), Kirchhoff's Voltage Law (KVL) statements and applications, Voltage Divider Rule (VDR), validity of source connections	Lecture 6
	Quiz 1	
7	Nodal Analysis technique, systematic node voltage formulation for independent sources, solving for branch currents and power dissipation with comprehensive examples	Lecture 7
8	Nodal Analysis with dependent sources, floating voltage source challenges, Supernode formulation and constraint equations	Lecture 8
9	Mesh Analysis technique, systematic mesh current formulation for independent sources, Supermesh formulation with current sources, solving for loop currents and element voltages	Lecture 9
10	Linearity of voltage and current, non-linearity of power, Superposition Theorem for DC circuits with independent and dependent sources, pre-midterm problem solving	Lecture 10
	Quiz 2	
	Midterm	

Lesson Plan - 2/2

11	Network reduction, Thevenin's Theorem derivation, motivation, and step-by-step determination of Thevenin equivalent voltage and resistance	Lecture 11
12	Norton's Theorem derivation and dual relationship to Thevenin's Theorem, determination of Norton equivalent current and resistance using test source methods	Lecture 12
13	Extraction of two-terminal I-V curves, inverse circuit design from graphical I-V responses, equivalent terminal representation of arbitrary linear networks	Lecture 13
14	Application of Thevenin and Norton theorems in complex networks, Maximum Power Transfer Theorem derivation, load matching conditions, efficiency considerations	Lecture 14
15	Energy storage elements, physical construction and terminal equations of capacitors and inductors, SI units, introduction to dynamic circuit response	Lecture 15
Quiz 3		
16	Source-free response of first-order RC circuits, time constant analysis, differential equation formulation, calculating capacitor voltage and current waveforms	Lecture 16
17	Step response of first-order RC circuits, transient and steady-state components, switch action analysis	Lecture 17
18	Source-free and step response of first-order RL circuits, time constant analysis, differential equation formulation, calculating inductor current and voltage waveforms	Lecture 18
19	Complex number representation, sinusoidal sources, frequency, amplitude, phase angle, Root Mean Square (RMS) voltage and current, dynamic AC signal visualization	Lecture 19
20	Concept of complex impedance and admittance, passive element response in the frequency domain, phasor analysis of AC circuits, instantaneous and average power	Lecture 20
Quiz 4		
21	Multi-frequency AC circuit analysis using the superposition theorem, comprehensive course review, synthesis of dynamic network analysis techniques	Lecture 21
Final Exam		

Course Materials

- Text and reference books

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Fundamentals of Electric Circuits	Charles K. Alexander, Matthew N. O. Sadiku	2019	6th	McGraw Hill Education	978-9353165505
2	Introductory Circuit Analysis	Robert L. Boylestad	2013	12th	Pearson Education India	978-9332518612
3	Electric Circuits	James W. Nilsson Susan A. Riedel	2010	9th	Pearson College Div	978-0136114994

- Other materials
  - Lecture Slides
  - Practice Problems
  - Previous Semesters' Mid and Final Questions
  - Video Lectures
  - Suggested Problems
  - Lab Sheets and Manuals
  - Lab Recordings

Mark Distribution††

Assessment	Criteria	Weightage (%)
Attendance	≥ 70% to attend final exam	5
Assignment	2 to 4	5
Quiz	Best 3 out of 4	15
Midterm	Central Exam	25
Final	Central Exam	25
Lab	Hardware Experiments	25
Total		100

Assessment	Criteria	Weightage (%)
Attendance	≥ 90% to attend final exam	2
Lab performance	6 experiments	5
Lab reports	2 Hardware	6
Lab tests	2 Hardware	12
Total		25

†† Mark distribution may change depending on the number of classes and syllabus for a semester.

Contact

Faculty Information

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Contact

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Consultation Hours†

- Sunday, Tuesday: 3:30 PM – 4:50 PM
- Monday, Wednesday: 2:00 PM – 3:20 PM
- Thursday: Consultation by Appointment

† Consultation is not makeup of any class. Specific concepts/theories/analytical problems/issues may be discussed.

Thank you for your attention